

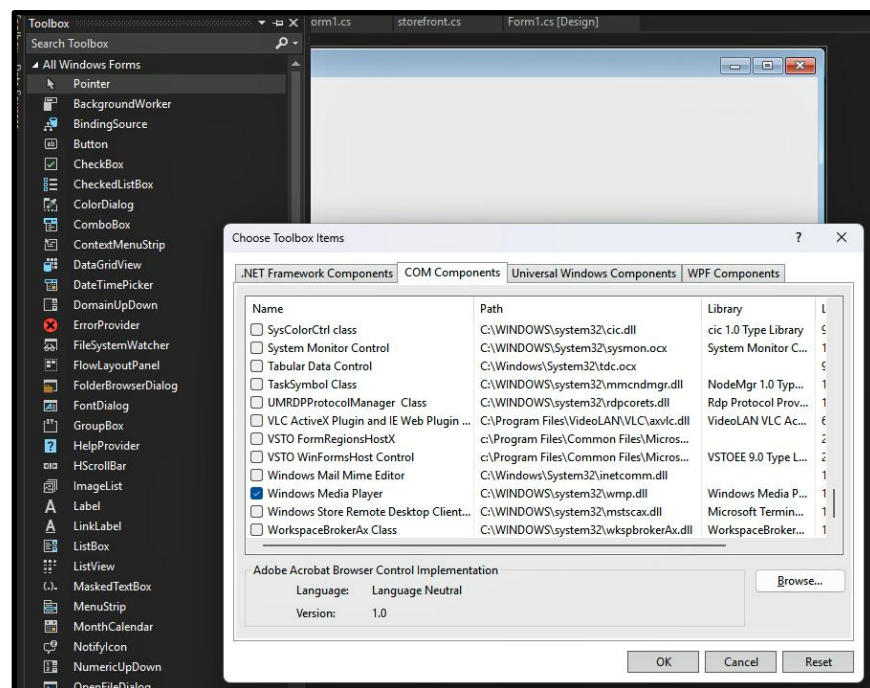
Chapter 15B: Storefront

We are now ready to start working on our CarDetails page. As mentioned in our last lesson this page is going to display images and videos of our buyer's car of interest. No problem, right? Just instantiate the VideoBox class. Certainly, Microsoft would have created one to correspond with its PictureBox class, right? Wrong, no VideoBox class in C#. Instead, you have to fall back onto the old Windows Media Player COM object.

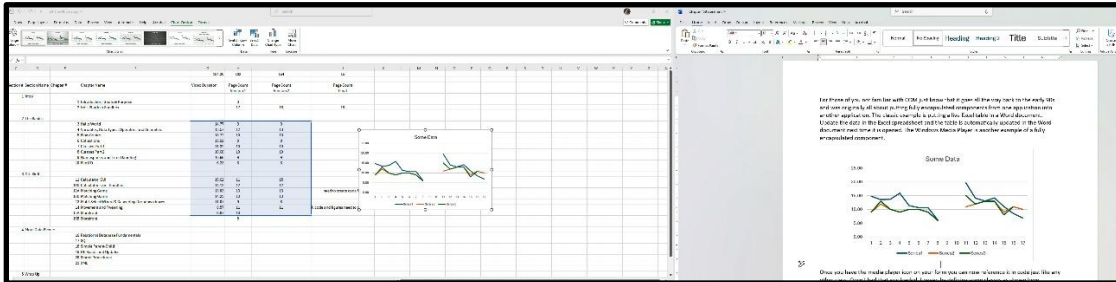
To make the Windows Media Player behave like a real class Visual Studio places a wrapper around the player thus giving it most of the properties and methods common to real classes. Unfortunately, to get that wrapper you need to add the player via the design environment. Given this restriction, we will first learn how to add a COM object to a form via the designer.

By default, COM objects are not listed in the Visual Studio Toolbox. To add them to the ToolBox open the ToolBox and right click. From the top menu select the Choose Items option. A wizard similar to that shown here will be displayed. (Figure 15B.1) Go ahead and click the COM Components tab and scroll down to the bottom to select the Windows Media Player option. Now simply click the OK button to add this COM object to your project.

Figure 15B.1: Windows Media Player



For those of you not familiar with COM just know that it goes all the way back to the early 90s and was originally all about putting fully encapsulated components from one application into another application. The classic example is putting a live Excel table in a Word document. Update the data in the Excel spreadsheet and the table is automatically updated in the Word document next time it is opened. The Windows Media Player is another example of a fully encapsulated component.



Once you have the media player icon on your form you can now reference it in code just like any other class. Once I had that guy loaded, I began by defining some objects to go on that form as shown here. (Figure 15B.2) Nothing interesting going on here but we will need these guys going forward.

Figure 15B.2: Class definitions

```
5 references
public partial class CarDetails : Form
{
    car my_car = new car();

    Font my_font = new Font(FontFamily.GenericMonospace, 10, FontStyle.Regular);
    Font my_font2 = new Font(FontFamily.GenericMonospace, 14, FontStyle.Bold);

    PictureBox pb1 = new PictureBox();
    PictureBox pic1 = new PictureBox();
    PictureBox pic2 = new PictureBox();
    PictureBox pic3 = new PictureBox();
    PictureBox pic4 = new PictureBox();

    PictureBox thumb1 = new PictureBox();
    PictureBox thumb2 = new PictureBox();
    PictureBox thumb3 = new PictureBox();
    PictureBox thumb4 = new PictureBox();

    1 reference
    public CarDetails(car this_car)
    {
        InitializeComponent();
        my_car = this_car;
    }

    1 reference
    private void CarDetails_Load(object sender, EventArgs e)
    {
        this.Size = new System.Drawing.Size(1200, 800);
        this.Text = "Car Details" + my_car.Manufacturer + " " + my_car.Model;
    }
}
```

Next it was time to recreate that logo. The AddLogo routine was simply copied from the home page with the addition of a Click event handler. It is common practice that when the user clicks the logo on any page other than the home page, they are returned to the home page of that site. That is of course what our LogoClick routine does for us here. (Figure 15B.3)

Figure 15B.3: LogoClick

```
1 reference
private void CarDetails_Load(object sender, EventArgs e)
{
    this.Size = new System.Drawing.Size(1200, 800);
    this.Text = "Car Details" + my_car.Manufacturer + " " + my_car.Model;

    // build GUI
    AddLogo();
}

1 reference
public void AddLogo()
{
    string my_logo = @"F:\Documents\C# The Whole Story\Section 3 GUI Stuff\Chapter 15A storefront\images\logo.jpg";
    if (File.Exists(my_logo))
    {
        Image logo = Image.FromFile(my_logo);
        PictureBox lg = new PictureBox();
        lg.Image = logo;
        lg.Size = new System.Drawing.Size(325, 150);
        lg.Location = new System.Drawing.Point(0, 0);
        lg.SizeMode = PictureBoxSizeMode.StretchImage;
        lg.Click += new EventHandler(LogoClick);
        this.Controls.Add(lg);
    }
}

1 reference
private void LogoClick(object sender, EventArgs e)
{
    Form1 form1 = new Form1();
    this.Hide();
    form1.ShowDialog();
    this.Close();
}
```

I next wanted to start setting up the details page. At the top center there will be a picture or video of the car in question while below will be the description. On the left-hand side will be up to four additional pictures of the car while on the right-hand side there will be up to four video thumbnails. My initial cut at the AddDetails routine is shown here. (Figure 15B.4)

Figure 15B.4: AddDetails

```
1 reference
private void CarDetails_Load(object sender, EventArgs e)
{
    this.Size = new System.Drawing.Size(1200, 800);
    this.Text = "Car Details" + my_car.Manufacturer + " " + my_car.Model;

    // build GUI
    AddLogo();
    AddDetails();
}

1 reference
private void AddDetails()
{
    // add the big image in the top center
    Image img1 = Image.FromFile(my_car.pics[0].filename);
    pb1.Image = img1;
    pb1.Location = new System.Drawing.Point(400, 150);
    pb1.Size = new System.Drawing.Size(450, 250);
    pb1.SizeMode = PictureBoxSizeMode.StretchImage;
    this.Controls.Add(pb1);

    // we have already added media player via designer so now we can reference it as a regular class object
    axWindowsMediaPlayer1.Size = new System.Drawing.Size(450, 250);
    axWindowsMediaPlayer1.Location = new System.Drawing.Point(400, 150);
    axWindowsMediaPlayer1.Visible = false;

    // AddThumbNails;
    // AddDescriptors;

    // add the I want to buy it button
    Button buy_it = new Button();
    buy_it.Text = "I want it!!";
    buy_it.BackColor = System.Drawing.Color.LightPink;
    buy_it.ForeColor = System.Drawing.Color.Black;
    buy_it.Location = new System.Drawing.Point(525, 650);
    buy_it.Size = new System.Drawing.Size(150, 50);
    buy_it.Font = my_font;
    // buy_it.Click += new EventHandler(ShowBuyACar);
    this.Controls.Add(buy_it);
}

1 reference
```

I next knocked out that AddThumbNails routine. Lots of repetition here so be sure to use your copy and paste keys carefully. (Figures 15B.5 & 6)

Figure 15B.5: AddThumbNails Part 1

```
1 reference
private void AddThumbNails()
{
    Label l1 = new Label();
    l1.Font = my_font;
    l1.Location = new System.Drawing.Point(275, 150);
    l1.Text = "Pics";
    this.Controls.Add(l1);

    Label l2 = new Label();
    l2.Font = my_font;
    l2.Location = new System.Drawing.Point(925, 150);
    l2.Text = "Vids";
    this.Controls.Add(l2);

    pic1.Location = new System.Drawing.Point(275, 175);
    pic1.Size = new System.Drawing.Size(75, 50);
    pic1.SizeMode = PictureBoxSizeMode.StretchImage;
    if (my_car.pics[1] != null)
    {
        Image img2 = Image.FromFile(my_car.pics[1].filename);
        pic1.Image = img2;
    }
    else pic1.Visible = false;
    // pic1.Click += new EventHandler(ShowNewImage);
    this.Controls.Add(pic1);

    pic2.Location = new System.Drawing.Point(275, 250);
    pic2.Size = new System.Drawing.Size(75, 50);
    pic2.SizeMode = PictureBoxSizeMode.StretchImage;
    if (my_car.pics[2] != null)
    {
        Image img3 = Image.FromFile(my_car.pics[2].filename);
        pic2.Image = img3;
    }
    else pic2.Visible = false;
    // pic2.Click += new EventHandler(ShowNewImage);
    this.Controls.Add(pic2);

    pic3.Location = new System.Drawing.Point(275, 325);
    pic3.Size = new System.Drawing.Size(75, 50);
    pic3.SizeMode = PictureBoxSizeMode.StretchImage;
    if (my_car.pics[3] != null)
    {
        Image img4 = Image.FromFile(my_car.pics[3].filename);
        pic3.Image = img4;
    }
    else pic3.Visible = false;
    // pic3.Click += new EventHandler(ShowNewImage);
    this.Controls.Add(pic3);

    pic4.Location = new System.Drawing.Point(275, 400);
    pic4.Size = new System.Drawing.Size(75, 50);
    pic4.SizeMode = PictureBoxSizeMode.StretchImage;
    if (my_car.pics[4] != null)
    {
        Image img4 = Image.FromFile(my_car.pics[4].filename);
        pic4.Image = img4;
    }
    else pic4.Visible = false;
    // pic4.Click += new EventHandler(ShowNewImage);
    this.Controls.Add(pic4);

    thumb1.Location = new System.Drawing.Point(925, 175);
```


Figure 15B.6: AddThumbNails Part 2

```
pic4.Location = new System.Drawing.Point(275, 400);
pic4.Size = new System.Drawing.Size(75, 50);
pic4.SizeMode = PictureBoxSizeMode.StretchImage;
if (my_car.pics[4] != null)
{
    Image img4 = Image.FromFile(my_car.pics[4].filename);
    pic4.Image = img4;
}
else pic4.Visible = false;
// pic4.Click += new EventHandler(ShowNewImage);
this.Controls.Add(pic4);

thumb1.Location = new System.Drawing.Point(925, 175);
thumb1.Size = new System.Drawing.Size(75, 50);
thumb1.SizeMode = PictureBoxSizeMode.StretchImage;
if (my_car.vids[0] != null)
{
    Image img5 = Image.FromFile(my_car.vids[0].thumbnail);
    thumb1.Image = img5;
}
else thumb1.Visible = false;
//thumb1.Click += (sender2, e2) => PlayVideo(sender2, e2, 0);
this.Controls.Add(thumb1);

thumb2.Location = new System.Drawing.Point(925, 250);
thumb2.Size = new System.Drawing.Size(75, 50);
thumb2.SizeMode = PictureBoxSizeMode.StretchImage;
if (my_car.vids[1] != null)
{
    Image img6 = Image.FromFile(my_car.vids[1].thumbnail);
    thumb2.Image = img6;
}
else thumb2.Visible = false;
//thumb2.Click += (sender2, e2) => PlayVideo(sender2, e2, 1);
this.Controls.Add(thumb2);

thumb3.Location = new System.Drawing.Point(925, 325);
thumb3.Size = new System.Drawing.Size(75, 50);
thumb3.SizeMode = PictureBoxSizeMode.StretchImage;
if (my_car.vids[2] != null)
{
    Image img7 = Image.FromFile(my_car.vids[2].thumbnail);
    thumb3.Image = img7;
}
else thumb3.Visible = false;
//thumb3.Click += (sender2, e2) => PlayVideo(sender2, e2, 2);
this.Controls.Add(thumb3);

thumb4.Location = new System.Drawing.Point(925, 400);
thumb4.Size = new System.Drawing.Size(75, 50);
thumb4.SizeMode = PictureBoxSizeMode.StretchImage;
if (my_car.vids[3] != null)
{
    Image img8 = Image.FromFile(my_car.vids[3].thumbnail);
    thumb4.Image = img8;
}
else thumb4.Visible = false;
//thumb4.Click += (sender2, e2) => PlayVideo(sender2, e2, 3);
this.Controls.Add(thumb4);
}
```

The AddDescriptors routine should be very easy for you by now. The only interesting thing is that Enabled method on the TextBox which precludes the user from updating any of the associated information. (Figure 15B.7)

Figure 15B.7: AddDescriptors

```
1 reference
private void AddDescriptors()
{
    Label manufacturer = new Label();
    manufacturer.Font = my_font2;
    manufacturer.Text = "Manufacturer: ";
    manufacturer.Location = new System.Drawing.Point(200, 450);
    manufacturer.AutoSize = true;
    this.Controls.Add(manufacturer);

    Label manufacturer2 = new Label();
    manufacturer2.Font = my_font2;
    manufacturer2.Text = my_car.Manufacturer;
    manufacturer2.Location = new System.Drawing.Point(400, 450);
    manufacturer2.AutoSize = true;
    this.Controls.Add(manufacturer2);

    Label model = new Label();
    model.Font = my_font2;
    model.Text = "Model: ";
    model.Location = new System.Drawing.Point(200, 480);
    model.AutoSize = true;
    this.Controls.Add(model);

    Label model2 = new Label();
    model2.Font = my_font2;
    model2.Text = my_car.Model;
    model2.Location = new System.Drawing.Point(400, 480);
    model2.AutoSize = true;
    this.Controls.Add(model2);

    TextBox textBox = new TextBox();
    textBox.Font = my_font2;
    textBox.Text = my_car.description;
    textBox.Location = new System.Drawing.Point(200, 510);
    textBox.Size = new System.Drawing.Size(850, 200);
    textBox.AutoSize = true;
    textBox.Multiline = true;
    textBox.Enabled = false;
    this.Controls.Add(textBox);
}
```

Now at this point some of you are inevitably saying “Rick, man your code is really simplistic.” To which I respond, “thank you.” Simple code is quick to write and easy to maintain. Gang, I intentionally write simple code whenever possible. There are plenty of opportunities to write the hard stuff but remember when you do that you will probably have to maintain that same code yourself because the more junior programmers are not going to be able to understand it.

The ShowNewImage event handler can now be uncommented and coded. Short little routine but this one is definitely not trivial. A lot going on in those few lines. (Figure 15B.8)

Figure 15B.8: ShowNewImage

```
4 references
private void ShowNewImage (object sender, EventArgs e)
{
    pbl.Visible = true;
    axWindowsMediaPlayer1.Visible = false;

    // stop video if playing
    if (axWindowsMediaPlayer1.playState == WMPLib.WMPPlayState.wmppsPlaying) axWindowsMediaPlayer1.Ctlcontrols.stop();

    // trade current pic for the one clicked
    Image temp = pbl.Image;
    PictureBox which_thumbnail = (PictureBox)sender;
    pbl.Image = which_thumbnail.Image;
    which_thumbnail.Image = temp;
}
1 reference
```

The PlayVideo routine is equally obscure, why Microsoft doesn't have a VideoBox class I have no idea. Using a 20+ year old video player is just ridiculous. Anyway, here is my cut at the routine. (Figure 15B.9)

Figure 15B.9: PlayVideo

```
4 references
private void PlayVideo(object sender, EventArgs e, int which_video)
{
    pbl.Visible = false;
    axWindowsMediaPlayer1.Visible = true;

    if (axWindowsMediaPlayer1.playState == WMPLib.WMPPlayState.wmppsPlaying) axWindowsMediaPlayer1.Ctlcontrols.stop();
    axWindowsMediaPlayer1.URL = my_car.vids[which_video].filename;
}
1 reference
```

We conclude our storefront example with a little humor on the buy_it click event handler. My ShowBuyACar routine is real straightforward. (Figure 15B.10) All it does is open a page called BuyACar and close the current page.

Figure 15B.10: ShowBuyACar

```
1 reference
private void ShowBuyACar(object sender, EventArgs e)
{
    this.Hide();
    BuyACar newPage2 = new BuyACar();
    newPage2.ShowDialog();
    this.Close();
}
1 reference
```

The Load event for the BuyACar Form is equally straightforward. Put up a PictureBox and set it's Image property to a woman scolding you for attempting to purchase an outrageously expensive car. (Figure 15B.11 and Figure 15B.12) Who says programmers don't have a sense of humor?

Figure 15B.11: BuyACar_Load

```
1 reference
private void BuyACar_Load(object sender, EventArgs e)
{
    this.Size = new System.Drawing.Size(900, 600);
    PictureBox pb1 = new PictureBox();
    pb1.Size = new System.Drawing.Size(850, 550);
    pb1.Location = new System.Drawing.Point(25, 25);
    pb1.SizeMode = PictureBoxSizeMode.AutoSize;

    Image i1 = Image.FromFile(@"F:\Documents\CS The Whole Story\Section 3 GUI Stuff\Chapter 15A storefront\images\scolding lady.jpg");
    pb1.Image = i1;
    this.Controls.Add(pb1);
}
```

Figure 15B.12: Scolding Lady



Okay, let's jump to the outside and talk about what we learned in this example.

- That the simple principles we learned about in those initial chapters can be combined to create some pretty amazing applications.
- That simple code is preferred whenever possible.
- That nesting TableLayoutPanel is probably not a great idea because managing their control collections can get pretty confusing.
- And most importantly, that programmers can have a sense of humor.